LLM Decoding Parameter & Sampling Analysis Report

1. Introduction & Objective

This project investigates how decoding parameters such as Temperature, Top-K, and Top-P significantly impact the consistency, diversity, and factual accuracy of Large Language Model (LLM) outputs.

2. Experimental Setup & Libraries

We utilized local models including Llama 3.1 and Qwen 2.5 via Ollama. The tech stack and libraries used include Python, Pandas, NumPy, Scikit-learn (for TF-IDF cosine similarity/diversity metrics), Streamlit (for the web interface), and FPDF.

3. Methodology & Evaluation

A benchmark dataset comprising factual, reasoning, and general prompts was tested across multiple runs. We evaluated output determinism (exact match consistency across runs) and lexical diversity using vector embeddings.

4. Results & Findings

- Temperature = 0.0 yields highly deterministic, consistent, and greedy responses.
- Higher temperatures (e.g., 0.8 - 1.2) increase lexical diversity and creativity, but elevate the risk of hallucinations on factual queries.

5. Conclusion

Decoding parameters allow fine-tuning the trade-off between strict determinism and creative variance depending on the target application.